

Software Engineering Department
SWER312 Software Testing and Quality Assurance
Fall 2025

Selenium WebDriver Project Assignment

Apply your knowledge of Selenium WebDriver and automated testing design patterns to create a project that includes various testing techniques.

Requirements:

1. **Automated Testing Design Pattern:**
 - Choose one of the automated testing design patterns (e.g., Page Object Model - POM).
 - Implement the chosen design pattern in your Selenium WebDriver project.
2. **BDD Framework (Cucumber Integration):**
 - Implement Behavior-Driven Development (BDD) using Cucumber.
 - Write Gherkin feature files that describe test scenarios in a readable “Given-When-Then” format.
 - Link feature files with step definitions implemented in Java and executed through Selenium WebDriver
3. **Test Cases:**
 - Implement at least 10 BDD test scenarios for the website you've already implemented.
 - Ensure that your test cases cover various functionalities of the website.
4. **Testing Techniques:**
 - Apply Cross-Browser Testing:
 - Test your application on different browsers (e.g., Chrome, Firefox, and Edge) to ensure cross-browser compatibility.
 - Apply Parallel Testing:
 - Run your test cases in parallel to optimize testing time.
 - Apply Data-Driven Testing:
 - Use external data sources (e.g., Excel, CSV) to drive your test cases with different sets of data.
 - Use Selenium Waiting Strategies:
 - Implement appropriate waiting strategies to handle dynamic elements or asynchronous behavior in your application.
5. **Group Work:**
 - You can work in groups of two students.
6. **Submission Deadline:**
 - The deadline for submission is **20/11/2025**.

Submission:

- Submit your project code along with a brief documentation highlighting the implemented design pattern, test cases, and testing techniques.
- Be prepared to discuss your project.